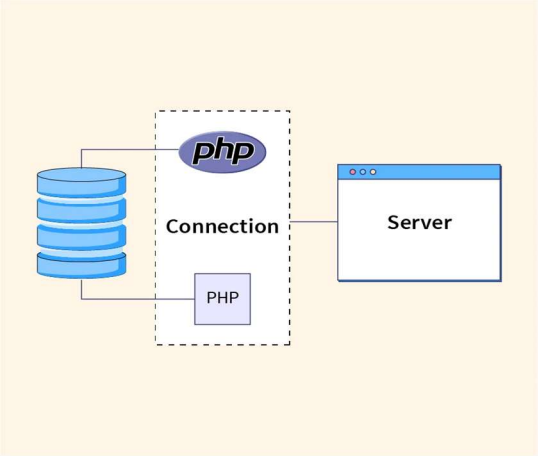
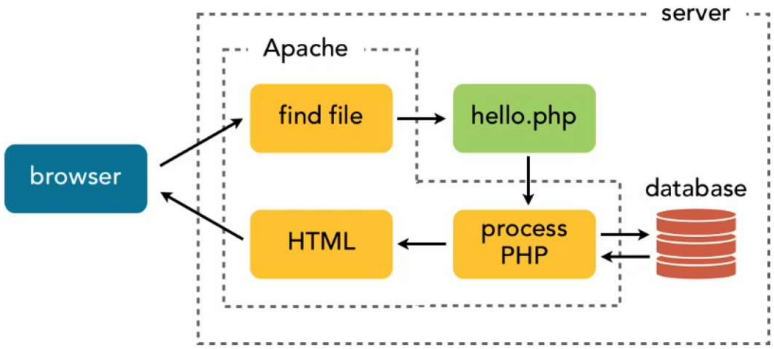
Full Stack Web Development Step by Step Notes





Real-Time Applications

- Student Management System
- Login & Authentication
- Feedback Form
- Online Registration System

Web development involves creating websites using frontend and backend technologies. The main technologies include HTML, CSS, JavaScript, PHP, and MySQL.

HTML is used to create the structure of a webpage. It defines elements such as headings, paragraphs, images, and forms. A basic HTML page includes a doctype declaration, html tag, head section with title, and body section where content is displayed.

CSS is used to style the webpage. It controls colors, fonts, spacing, and layout. CSS can be applied inline, internally, or externally. Modern layouts use concepts like flexbox and grid to create responsive designs.

JavaScript is used to add interactivity to webpages. It allows handling user actions such as button clicks, form validation, and dynamic content updates. It works with the Document Object Model to manipulate HTML elements.

PHP is a server-side scripting language used to process data. It handles form submissions, performs calculations, and interacts with databases. PHP code is executed on the server and the output is sent to the browser.

MySQL is a relational database management system used to store data. Data is organized into tables consisting of rows and columns. SQL commands are used to create databases, create tables, insert data, and retrieve data.

To create a database, first use the command CREATE DATABASE followed by the database name. Then select the database using USE command. Tables are created using CREATE TABLE with fields like id, name, and email.

To connect PHP with MySQL, a connection is established using mysqli_connect function with parameters such as host, username, password, and database name. If the connection fails, an error message is displayed.

Data can be inserted into the database using an HTML form. The form sends data to a PHP

file using POST method. The PHP file retrieves the data using $_POST and inserts it into the database using an SQL INSERT query.

Data can be displayed by retrieving records from the database using SELECT query. The results are fetched using mysqli_fetch_assoc and displayed on the webpage.

For security, prepared statements should be used instead of direct queries. Prepared statements help prevent SQL injection by binding user inputs safely.

A simple project can be built as a student registration system where users can enter their name and email, store it in the database, and view the stored records.

The learning flow should start from HTML, followed by CSS, JavaScript, PHP, and finally MySQL. After learning each component, they should be integrated to build a complete application.

Students should practice by creating forms, connecting to databases, inserting and displaying data, and gradually implementing update and delete operations.

HTML (Structure)

HTML defines the structure of web pages.

Basic Code:

```
<!DOCTYPE html>
<html>
<head>
<title>Page</title>
</head>
<body>
<h1>Welcome</h1>
<p>Text</p>
</body>
</html>
```

Diagram:
User -> Browser -> HTML Structure -> Display

CSS (Styling)

CSS is used to design the webpage.

Example:

```
body { background-color: lightblue; }
h1 { color: blue; }
```

Diagram:
HTML -> CSS Applied -> Styled Output

JavaScript (Interactivity)

JavaScript adds dynamic behavior.

Example:

```
<button onclick="show()">Click</button>
<script>
function show(){ alert("Hello"); }
</script>
```

Diagram:
User Action -> JS Event -> DOM Update

PHP (Backend)

PHP processes data on the server.

Example:

```
<?php echo "Hello"; ?>
```

Diagram:
Client Request -> Server (PHP) -> Response

MySQL (Database)

MySQL stores data in tables.

Example:

```
CREATE DATABASE student_db;
CREATE TABLE students(id INT, name VARCHAR(50));
```

Diagram:
PHP -> MySQL -> Data Storage

Integration Flow

Complete Flow:

User -> HTML Form -> PHP -> MySQL -> PHP -> Browser

Code Integration Example:

HTML:

```
<form method="POST" action="insert.php">
<input type="text" name="name">
<input type="submit">
</form>
```

PHP:

```
<?php
$conn = mysqli_connect("localhost","root","","student_db");
$name = $_POST['name'];
mysqli_query($conn,"INSERT INTO students(name) VALUES('$name')");
?>
```